

Positive Literature Answer to the L_2 Nonnegative Schauder-Basis Question

FA/Banach run packet

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Status

This is a literature-already-answered packet for a partial subcase. It records that the $p = 2$ case of a nonnegative Schauder-basis question mentioned by Nikolski–Volberg [1] was answered positively by Freeman–Powell–Taylor [2]. It is not an original proof from this run.

Original Problem

Nikolski–Volberg [1] construct a normalized uniformly minimal complete sequence of nonnegative functions in $L^2_{\mathbb{R}}(0, 1)$, and observe that this sequence is not a basis. Immediately afterward they write that the existence of a nonnegative Schauder basis in L^p , $p > 1$, is open to their knowledge, while the case $p = 1$ is known by Johnson–Schechtman.

Supporting Answer

Freeman–Powell–Taylor [2] return directly to this problem. Their introduction says that, after the Johnson–Schechtman L_1 construction, the conditional positive Schauder-basis problem remained open for every $1 < p < \infty$. Their main theorem in Section 2 proves:

Literature Answer. *For every $\varepsilon > 0$, $L_2(\mathbb{R})$ has a Schauder basis consisting of nonnegative functions, with basis constant at most $1 + \varepsilon$.*

The same paper remarks that classification theorems extend this conclusion to all separable $L_2(\mu)$ spaces. Therefore the Hilbert-space case of the Nikolski–Volberg $p > 1$ question has a positive literature answer.

Scope Limitations

This packet does not claim a full solution for all $p > 1$. Freeman–Powell–Taylor explicitly leave the remaining cases $1 < p < \infty$, $p \neq 2$, open as Problem 5.1. They do prove, for all $1 < p < \infty$, a weaker positive result: there is a nonnegative Schauder basic sequence whose closed span contains an isomorphic copy of $L_p(\mathbb{R})$.

Verification Notes

The original source and the supporting source are distinct. The supporting paper is a later work whose title and Theorem 2.2 address exactly the nonnegative Schauder-basis problem in the L_2 case. The local duplicate search found a failed attempt on separate Markushevich-basis questions from [2], but no packet recording this partial literature answer to [1].

References

- [1] N. Nikolski and A. Volberg, *On the Sign Distributions of Hilbert Space Frames*, arXiv:1812.06313.
- [2] D. Freeman, A. M. Powell, and M. A. Taylor, *A Schauder basis for L_2 consisting of non-negative functions*, arXiv:2003.09576; Math. Ann. 383 (2022), 181–208.